

SEMP

Generic Application
Digital Interlocking OBB

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LOG OF CHANGES			
Revision	Date	Author	Change Description
01 PD 01	2025-03-24	E. Stemmer	First proposal for review
01 RL	2025-04-03	E. Stemmer	First Edition for SOR milestone 8.4.2025
02 PD 01	2025-10-02	E. Stemmer	Proposal for review – update 4 th offer & SRR-1 milestone
02 RL	2025-10-10	E. Stemmer	Second Edition for SRR-1 milestone (prior to contract)

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1. SCOPE

The System Engineering Management Plan (SEMP) describes the organisation, work packages, milestones and application of the Management System DDQS process release number 2025-08.

1.1 IDENTIFICATION

1.1.1 GENERAL

Table 1: Solution Identification

Project Name:	DSTW ÖBB Entwicklung
Project ID:	100025795 / ATRELDE91
Solution / System:	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) ¹
Contract:	Digitales Stellwerk DSTW – Rahmenvereinbarung ²
Identification³:	LockTrac 6131 ELEKTRA LockTrac 6192 DCA FieldTrac 6315 AzLM
Customer:	ÖBB
End Customer:	ÖBB
SIL Level:	SIL 4 ⁴

Table 2: Security / Maturity Level

Security Level:	SL-T 3, SL-C 3 ⁵
Maturity Level:	ML 3

This SEMF only deals with the scope of the price position Basic Virtualization (B-Vi) of the solution and contract mentioned in Table 1: Solution Identification.

¹ Please note that at this stage the focus of the development of the Generic Application is on Phase 1 and this is indicated by 1.0.0 – for the phases see 0

² Please note that this SEMF Edition is based on the 4th Edition of the offer placed to ÖBB at 12th of September 2025 and the negotiations process is ongoing. The KOM will take place after the contract has been signed. Current activities are based on the customer requirements [CRS]. SOR will be repeated after official KOM.

³ The identification refers to the key products/solutions included in the Digital Interlocking

⁴ On SIL level please refer also to Table 3: WBS managed as Solution

⁵ On the SL please note that ÖBB and also EULYNX clearly asks for SL3, an end-to-end implementation in Phase 1 will most likely NOT be possible. However, the setup and the describing documents shall be prepared for SL3

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“Basic virtualization” (B-Vi) is the minimum technical requirement to be implemented upon the first delivery of a DSTW including the implementation of all functions of the EULYNX specifications.

The operations control centres (OMC) will control the field elements connected via a network through central signalling applications (VSTWs) located in data centres (DC). The overall system includes the central computer centres with DSTW central units, the security services platform, the MDM, the IP network, the power supply, the object controllers and the corresponding outdoor equipment.

In the B-Vi variant, local redundancy of the DSTW central unit (DSTW-ZE) is required. This means that a redundant application of the interlocking logic (VSTW) must be operated on a second server hardware at the same computer centre location.

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1.1.2 WBS MANAGED AS SOLUTION

Table 3: WBS managed as Solution

WBS	Identification	SEM	SIL	Initial Product Version	Target Product Version	Target date
ATRELDE91-E0	Generic Application Digital Interlocking OBB	E. Stemmer	SIL4	NA	PV GA DIL OBB 1.0.0*	Milestone FQR, see section 1.3.1
ATRELDE91-E0.WP3.DEV.3.2.1 & ATRENDDB35-E1	Solution Element Virtual Interlocking Phase 1	P. Bargiel	SIL4	ELEKTRA PV GEN 2.19.0 / ELEKTRA BL OBB 29.30 ⁶	VIL OBB 1.0.0 (ELEKTRA PV GEN 2.20.0 / ELEKTRA BL OBB 40.00 / EEC SW OBB 04.00)	Milestone FQR, see section 1.3.1
ATRELDE91-E0.WP3.DEV.3.2.2	Solution Element IPS Solution Phase 1	V. Vasilev	BI ⁷	BFZ-3.0 (IPS 30.0.0, ADM 30.0.0)	IPSS OBB 1.0.0 (IPS 40.0.0 ADM 40.0.0 HW GENCAB SERV 1.0 HW GENCAB NETW 1.0)	Milestone FQR, see section 1.3.1
ATRELDE91-E0.WP3.DEV.3.2.3	Solution Element Digital Field Elements Phase 1	B. Mandl	SIL4	NA	DFE 1.0.0 (DFE GEN 1.0.0 DCA3.1.0 AzLM v7.10 LCSi Gateway 1.0) VCRM 1.0.0 (OEC SW 1.0.0 DCA 2.x IOC)	Milestone FQR, see section 1.3.1

* Note: This SEMP describes the first product version for the first stations/installations; to fulfill the complete set of customer requirements two additional product versions are planned and will be described in future versions of the SEMP

⁶ Please note that for the overall solution the planned feature content of ELEKTRA BL OBB 40.00 is following the future ELEKTRA 2 Baseline OBB 29.40

⁷ For IPS and ADM no safety case is planned, only a validation report will be available.

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1.1.3 WBS MANAGED AS WORK PACKAGES

A detailed view on all WBS elements is listed in section 3.6.

Table 4: WBS managed as Work Packages

Solution: ATRELDE91-E0					
WBS	Ident.	WPM	Initial Version	Target Version	Target date
ATRELDE91-E0.WP1 ⁸		H. Wolfram	NA		Milestone FQR, see section 1.3.1
ATRELDE91-E0.WP2		E. Stemmer	NA		Milestone FQR, see section 1.3.1
ATRELDE91-E0.WP4		NN (manager will be announced later)	NA		Milestone FQR, see section 1.3.1

⁸ See [PMP]

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1.2 SYSTEM OVERVIEW

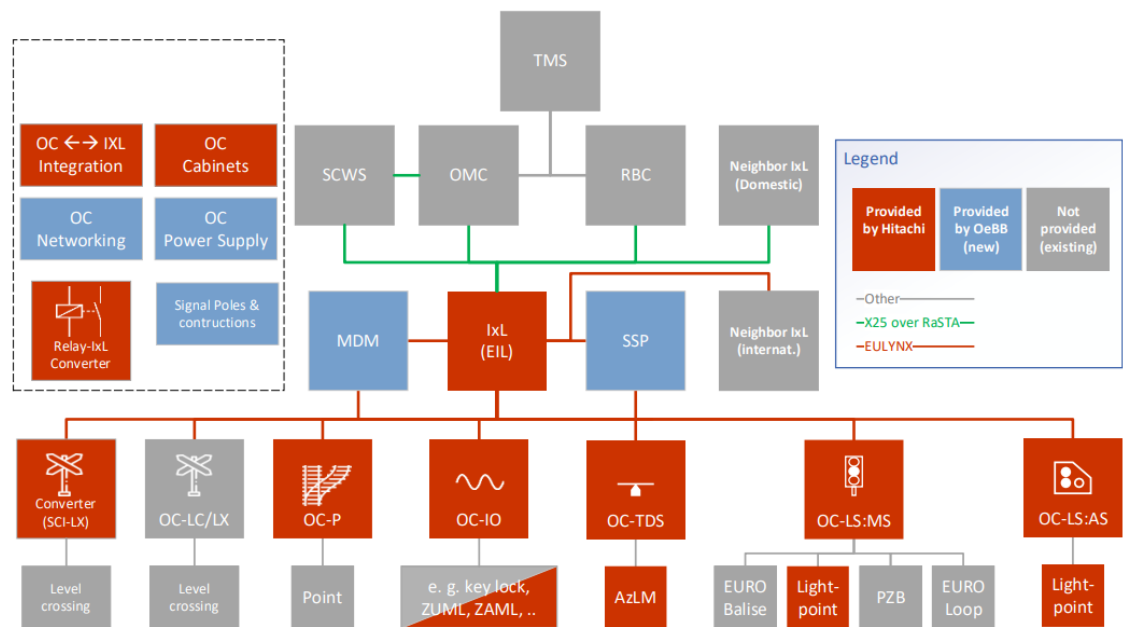
For an overview about customer expectation see:

[CRS] „OBB_RA-DSTW-D1_Leistungsbeschreibung“

The detailed tasks are listed in the document, but in summary it should be noted:

- 1. Development of the new ELEKTRA interlocking generation based on the ÖBB specifications for the digital interlocking solution and new field elements,
- 2. Possibility of up to 287 interlockings (IxL) virtualized at two to three centralized computing centres (incl. geo-redundancy),
- 3. reduction of facilities/buildings needs for the interlocking core and maintenance needs for the field elements by use of virtualization and object controllers,
- 4. address the European standardization efforts of railway operators, known as the “EULYNX initiative”,
- 5. cyber security on base of the IEC 62443, TS50701, EULYNX BL 4 and EU-Rail System Pillar, tackling the upcoming NIS2 directive and changed cyber threat situation,
- 6. Enabling roll-out projects,
- 7. Setting up a service infrastructure,
- 8. Overall integration of the interlocking solution into the ÖBB infrastructure

Figure 1: Scope Split



For more information about used products see [PBS]. For clarification on Figure 1: the lightpoints will be integrated and delivered with the rollout, however a new development is NOT included in the project scope.

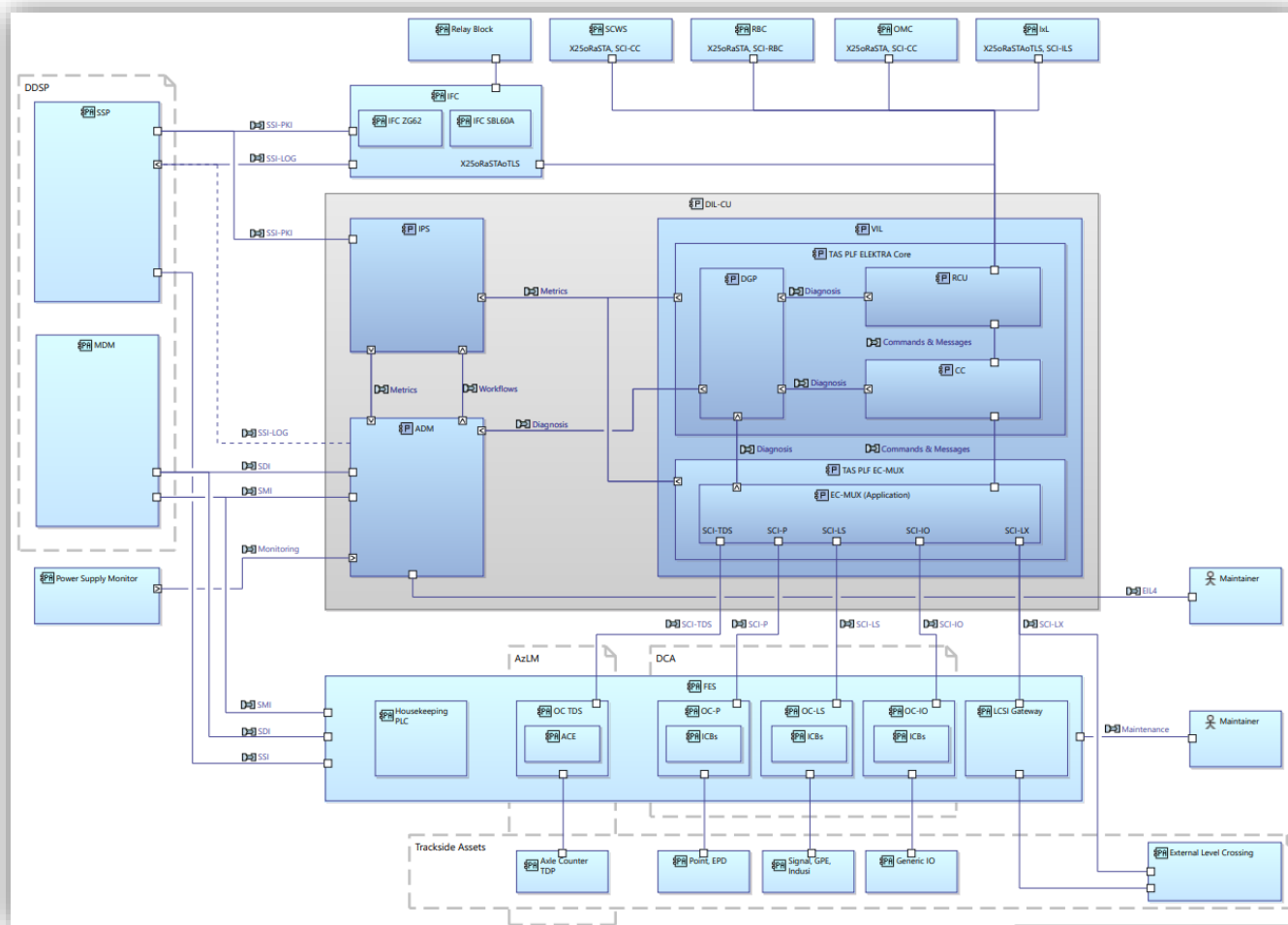
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Figure 2: System Architecture and Solutions



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1.3 KEY POINTS

1.3.1 MILESTONES FOR SOLUTION

Table 5: Milestones Generic Application Digital Interlocking OBB 1.0.0

Milestone	Target Date	Lead
Solution Orientation Review 1 (SOR-1)	9.4.2025	SEM
Solution Requirement Review 1 (SRR-1)	10.10.2025	SEM
<i>Hand-over Review & Kick-Off Meeting (KOM)</i>	x.y.2025	PM
Solution Orientation Review 2 (SOR-2)	<i>Right after KOM / Jan 2026</i>	SEM
Solution Requirement Review (SRR-2)	MS E03 ⁹ +3 months / 07/26	SEM
Solution Functional Review (SFR)	Partly agile approach ¹⁰ / MS E03 +5 months / 09/26	SEM
Preliminary Design Review (PDR)	Agile approach	SEM
Critical Design Review (CDR)	Partly agile approach / 31.1.2027	SEM
Test Readiness Review (TRR1)	15.12.2027 ¹¹	IVM
Technical Qualification Review 1 (TQR1)	31.3.2028	SEM
Test Readiness Review 2 (TRR2)	31.3.2028	SEM
Technical Qualification Review 2 (TQR2)	31.5.2028	SEM
Final Qualification Review (FQR)	31.7.2028 ¹²	SEM

- All dates are under the assumption of a contract signature by December 2025 (=T0)

The milestones are related to the content of the Phase 1 of deliverables as included in the offer submitted to ÖBB in September 2025 – for details see section 3.9.1.

⁹ MS E03 Pflichtenheft Freigabe from Customer ÖBB defined in Table 6. MS E03 is currently forecasted with April 2026 (under the assumption an agreement can be done with ÖBB for and early release of Pflichtenheft for Phase 1 only)

¹⁰ The Definition of Done (DoD) for the milestones marked as agile will be defined together with all three Solution Elements and will be along the definitions in [S25] & DoSD reviewed along [S30]Template of DoX Templates

¹¹ Please note that this milestone will be seen as Integration test start as not all Solution Elements will have reached full TQR1 at this stage – the detailed plan is under design and the approach is overlapping testing. Also a high dependency from the ÖBB inputs is clearly understood.

¹² The overall target is to have a first pilot in Operation by end 2028 – therefore the FQR date for the Generic Application is set latest mid of 2028 overlapping with first Rollout activities and generation of Specific application

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Table 6: Customer Milestones from ÖBB for Solution Development

Key Milestone	Name	Freigabe Strategie
MS E01	Auftragsvergabe	Vertragsunterschrift
MS E02	Entwicklungsplanung	Planungsdokumente freigegeben
MS E03	Pflichtenheft	Pflichtenheft Freigabe
MS E04	System Requirement Specification	Information an AG
MS E05	System Architectural Specification	Information an AG
MS E06	Testkonzept	Freigabe AG
MS E07	Planungshandbuch FES	Information an AG
MS E08	Planung /Design	Abschluss Phase 1
MS E09	Implementierung EULYNX und Lastenhefte	Information an AG
MS E10	Generische Tests	Information an AG
MS E11	System Integration	Information an AG
MS E12	Projektierungstool "spezifische Anwendungen"	Information an AG
MS E13	Umsetzung	Abschluss Phase 2
MS E14	Abnahme Generische Anwendung (FAT)	Erfolgreicher FAT abschluss
MS E15	Systemvalidierung	Information an AG
MS E16	Sicherheitsnachweis "Generische Anwendung"	Gutachten freigegeben
MS E17	Bereit zur Systemfreigabe "Generische Anwendung"	Gutachten freigegeben
MS E18	Übernahme	Abschluss Phase 3

The linking of Table 5: Milestones Generic Application Digital Interlocking OBB 1.0.0 and Table 6: Customer Milestones from ÖBB for Solution Development is visible in the [PMP] and Project Master Schedule as presented in the [EngDB].

1.3.2 DELIVERABLES

see Table 3.

1.4 RELATION WITH OTHER PLANS

See also Documentation Plan [DoP].

The SEMP describes the application of the Management System (Release from 2025-08) to the design, development and integration of the Generic Application Digital Interlocking OBB 1.0.0.

Table 7 : Subordinate Planning Documents

Title	Description	Ref.
<i>Project Management Plan</i>	Focuses on the Management System including organization, policies and plans that are carried out by Hitachi Rail AT in relation to the Project.	[PMP]

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Title	Description	Ref.
<i>Configuration Management Plan</i>	Provides details of the formal Configuration Management process.	[CMP]
<i>Quality Management Plan</i>	Describes requirements for all quality assurance measures and quality assurance activities to be carried out in the solution as contribution to the fulfilment of the project goals.	[QMP]
<i>RAM Plan</i>	Provides details of the tasks and activities about Reliability, Availability and Maintainability of the Solution. Project specific RAM Plans per solution element.	[RAMP]
<i>Safety Plan</i>	Provides details of the tasks and activities of Solution safety management and Solution safety engineering required to identify, evaluate and eliminate or control hazards throughout the life cycle. Reference to Project Specific Safety Plan	[PSP]
<i>IV Plan</i>	Is the general overview how it is planned to manage and test the Generic Application with the three Solution Elements. The project specific plan contains the ÖBB requirements and is named V&V plan	[IVP]
<i>VQ Plan</i>	Is the general overview how it is planned to manage and validate the Generic Application with the three Solution Elements. The generic VQP and a project specific plan following the ÖBB requirements is provided	[VQP]
<i>Security Management Plan</i>	Is the general overview how it is planned to address the security aspects in the Generic Application with the three Solution Elements. The generic SecMP and a project specific plan following the ÖBB requirements is provided	[SecMP]

1.5 BASELINES OF SOFTWARE CONFIGURATION ITEMS AND DOCUMENTATION LABELS

1.5.1 DOCUMENTATION LABELING

For documentation available via DWF following labels are used as defined also in the Documentation Plan [DoP]. The labelling shall follow the definition of Target Product Versions as per Table 3: WBS managed as Solution.

Details on the Product versions are given in the [CMP] and the solution Engineering Plans of the Solution Elements ([SEMP SE VIL], [SEMP SE DFE], [SEMP SE IPSS]).

For Documents on the Generic Application Level following labels shall be applied

- “DSTW” for all general documents related to the Digital Interlocking
- “PV_GA_DIL_OBB_1.0.0” for the documents which defines the first Baseline (Phase 1) of the Generic Application

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1.6 DOCUMENT VERSION HISTORY

1.6.1 EDITION 01

Initial edition.

1.6.2 EDITION 02

Update:

- Updates along 4th offer submitted to ÖBB 12th of September 2025
EULYNX BL 4R4
Security along System Pillar
no DDSP in Phase 1
no Retrofit
- Updated timeline / masterplan including changes done for submission to ÖBB along (see DVa)
new ÖBB requested timeline
TAS Platform updates
AzLM SOR – some milestones pending
DCA SOR – pending final milestones
updates in the overall project including dependencies and linking
- Document update for SRR-1
scope in section 3.9
timeline in section 1.3.1
referenced documents in section 2.1

1.7 OPEN POINTS

- SRR-1 milestones of the Solution Elements VIL/IPSS/DFE have been performed and the status was analysed with QAM/VAL/SAM and the requirements managers.

From this perspective and understanding that the SRR-1 is prior to the contract signature with the customer and therefore several questions which need to be agreed with the customer, the SEMP describes a status of the project with the intention to:

- analyze the actual status of the work with main focus on planning documentation (SEMP, WPDs, QMP, VQP, SP, ...)
- identify the chosen structure of requirements and specification document – including check of maintainability, completeness, efforts in Validation, ...
- identify gaps like how to deal with Industrialisation (see chapter 3.10)
- check on the agreed timeline and resources

having this in mind with SRR-1 a lot of reservations and open points will be identified which can be only solved during the next half year. With contract signature the “normal” DDQS will start with KOM/SOR-2/SRR-2/... - while the work in the project will proceed with all possible preparation work at this stage to come to a first pilot with ÖBB by YE 2028 along customer expectation and our offer

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- With the learning during SOR-1 and SRR-1 the point is raised to have a full time Architect on top to the PDA in the Generic Application as the approach with a team of Architects could be not sufficient. This need to be addressed and solved till SOR-2

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2. REFERENCES AND TERMS

2.1 REFERENCED DOCUMENTS

The relevant version of a document is referenced in the Documentation Plan [DoP]. Several planning documents are concurrent in progress until milestone SOR, therefore the edition 01 is referenced of Documentation Plan / Process Verification Report. Until milestone FQR the Documentation Plan / Process Verification Report for this Solution will be updated and tracked within the Version in TOL see TRN [DoP].

Table 8 : Referenced Documents

Id	Nr./Title	Reference	Version
[DoP]	Documentation Plan (DoP) AnvNr_3BU_17300_0000-9999_ELEKTRA_4.0	3BU 17300 1201 ADAPA	01PD02
[CRS]	DSTW Streckenausrüstung ÖBB - Teil D Leistungsbeschreibung und Lastenhefte	3BU 17300 0004 DSAPC	04
[PMP]	Project Management Plan	3BU 17300 1000 DPAPC	01PD02
[RiskReport]	Risk Report	100025795 ATRELDE91-E0 DSTW ÖBB Entwicklung Risk List 20250930 ED02.xlsx	02
[EngDB]	Engineering Dashboard (sharepoint)	100025795 ATRELDE91 DSTW-ÖBB- Entwicklung EngDB 202508.xlsm	August 2025
[QMP]	Quality Management Plan ELEKTRA	3BU 15000 5244 QMAPA	08PD01
	Quality Management Plan DSTW	3BU 17300 1202 QMAPC	01PD01
[SaCo]	Sicherheitskonzept, ELEKTRA	3BU 15000 0622 DUAPC	04
	Safety Concept - Two Diverse SW Channels on a Redundant Platform	3BU 15000 6957 DUAPA	01PD02
	Safety Concept Elektra 4 Communication	3BU 15000 6956 DUAPA	01PD05
	Safety concept DIL	3BU 17300 1212 DUAPC	SOR-2
[PSP]	Safety Plan DSTW	3BU 17300 1204 DUAPA	01
[SoC]	Scope of Change (SoC)	N/A	
[DVa]	Design Validation approval (DVa)	OeBB Bid DVa.docx	12
		extended with updates and details from "Digitales Stellwerk - Technische Konzepte" 3BU 17300 0010 DSAPC	04

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Id	Nr./Title	Reference	Version
[ToolClassSW]	Software Development, Tool Classification, EN 50128	3BU 15000 5135 AAAPA¹³	latest
[ToolClass]	Thales Austria Tool Classification EN50129	3BU 15000 5135 190EN	latest
[CMP]	Configuration Management Plan ELEKTRA DSTW CMP	3BU 15000 3338 QMAPA 3BU 17300 1210 DUAPA	11 SOR-2
[RMP]	Requirements Management Plan (RMP) Generic Application Digital Interlocking OBB	3BU 17300 1207 DUAPA	01
[RAMP]	RAM Plan Generic RAM Plan Specific Project specific RAM Plans per solution element	3BU 15000 1712 DUAPA 3BU 17300 1209 DUAPC 3BU 17300 1209 DUAPA 3BU 17300 2009 DUAPA	06 01 01 01
[MS-Report]	Milestone Report Solution	3BU 17300 1208 BCAPA	01PD02
[OBB_RA-DSTW]	OBB_RA-DSTW-D1_Leistungsbeschreibung	3BU 17300 0004 DSAPC	03
[IVP]	Integration and Verification Plan, ELEKTRA 2.x Project Specific Integration and Verification Plan (IVP)	3BU 15000 0654 DUAPA 3BU 17300 1206 DUAPC	03 01
[VQP]	Validation and Qualification Plan, ELEKTRA 2.x Project Specific Validation and Qualification Plan (VQP)	3BU 15000 0655 DUAPA 3BU 17300 1205 DUAPA	03 01
[SecMP]	Product Security Management Plan Elektra Project Specific Security Management Plan (SecMP)	3BU 15000 5300 DPAPA 3BU 17300 1213 DPAPA	03 01PD02
[PBS]	Product Breakdown Structure	3BU 17300 1215 DPAPA	02
[SEMP SE VIL]	System Engineering Management Plan Solution Element Virtual Interlocking	3BU 15000 6955 DPAPA	01PD02
[SEMP SE DFE]	SEMP Solution Element - Digital Field Elements	3BU 62800 0000 DPAPA	02

¹³ This document will need to be updated according to the successor [S2] EN 50716:2023 – Railway application – Requirements for software development of EN 50128

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Id	Nr./Title	Reference	Version
[SEMP SE IPSS]	System Engineering Management Plan Solution Element IPSS	<u>3BU 17300 2001 DPAPA</u>	01

2.2 PROCESS BASIS

The project is set up with the Management System, release number 2025-08 as process basis. Changes in this process basis (e.g. adaptation to a new release of Management System) have to be analysed, especially regarding the impact on already existing work products, and have to be documented and approved.

2.3 STANDARDS AND PROCEDURES

Table 9 : Standards and Procedures

Nr./Title	Publishing Date/Reference	Version/Revision
[S1] IRIS Certification rules:2020 based on ISO/TS 22163:2017	2017	n/a
[S2] EN 50716:2023 – Railway application – Requirements for software development	2023	n/a
[S3] EN 50129:2018 + AC:2019 Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signalling	2018/2019	n/a
[S4] ISO 9001	2015	n/a
[S5] EN 50126-1:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 1: Generic RAMS Process	2017	n/a
[S6] EN 50126-2:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 2: Systems Approach to Safety	2017	n/a
[S7] Management System Process Manage Risks and Opportunities	Part of 87200984-00-PRJ-GRP-EN (Chapter 8.6)	2025-08
[S8] Management System Process Manage Project	87200984-00-PRJ-GRP-EN	2025-08
[S9] Minimum Requirements for the configuration management implementation	Part of 87100077-CFG-GRP	007
[S10] Management System Process 6.1 DDQS-SW - Design develop and qualify the Software solution	Part of 87202125-00-DDQ-GRP-EN	2025-08
[S11] Management System Process 6.1 DDQS-HW - Design develop and qualify the Hardware solution	Part of 87202126-00-DDQ-GRP-EN	2025-08

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Nr./Title	Publishing Date/ Reference	Version/ Revision
[S12] Management System Process Validate the Solution	Part of 87202127-00-DDQ-GRP-EN	2025-08
[S13] Management System Process Integrate and Verify the Solution	Part of 87202127-00-DDQ-GRP-EN	2025-08
[S14] Management System Process Design the Solution	Part of 87202127-00-DDQ-GRP-EN	2025-08
[S15] Management System Process 9.7 CFG - Manage Configuration	87202424-00-CFG-GRP-EN	2025-08
[S16] Management System Process 9.3 Control Export	87299999-00-EXP-GRP-EN	2025-08
[S17] Management System Guarantee quality and manage continuous improvement	87200214-00-QCI-GRP-EN	2025-08
[S18] Protection of Group Information	87201725-GOV-GRP-EN	005
[S19] Personal Data Protection Policy	87212828-GOV-GRP-EN	001
[S20] Project Life Cycle	83530275-DDQ-AUT-EN	005
[S21] Document Management Rules	83530061-DOC-AUT-EN	013
[S22] Milestone Template Solutions	83530019-DDQ-AUT-EN	013
[S23] Document List and Review Matrix for engineering projects	83530224-DOC-AUT-EN	010
[S24] HSE Instruction	83530048-HSE-AUT-DE	020
[S25] Agile Definition SMP	83530312-DDQ-AUT-EN	002
[S26] ISA/IEC 62443 - Security for industrial automation and control systems IEC 62443-3-3: System security requirements and security levels, Edition 1, 09 2013	2013	01
[S27] ISA/IEC 62443 - Security for industrial automation and control systems IEC 62443-4-1: Secure product development life-cycle requirements, Edition 1, 01 2018	2018	01
[S28] ISA/IEC 62443 - Security for industrial automation and control systems IEC 62443-4-2: Technical security requirements for IACS components, Edition 1, 04 2019	2019	01
[S29] EN 50159, + A1:2020 Railway applications – Communication, signalling and processing systems – Safety-related communication in closed transmission systems	2020	n/a
[S30] Template of DoX Templates	83530313-DDQ-AUT-EN	002

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2.4 TERMS, ABBREVIATIONS AND ACRONYMS

With future releases and in the DIL project the intention is to use a global document for abbreviations. This is under establishment and can be found on sharepoint:

[Abbreviations for Glossary DIL.xlsx](#)

2.4.1 ROLE ABBREVIATIONS

Table 10 : Role abbreviations

ARC	Architect
CM	Configuration Manager
DM	Document Manager
HoD	Head of Discipline
IVM	Integration and Verification Manager
IVVQM	Integration, Verification, Validation and Qualification Manager
PDA	Project Design Authority
PLA	Product Line Architect
PLM	Product Line Manager
PSM	Project Security Manager
QAM	Quality Assurance Manager
RAM-AM	Reliability, Availability, Maintainability Assurance Manager
RE	Requirements Engineer
RM	Requirements Manager
TM	Test Manager
SAM	Safety Assurance Manager
SecAM	Security Assurance Manager
SEM	Solution Engineering Manager
SM	Senior Management
VAL	Validator
VER	Verifier
WPM	Work Package Manager

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2.4.2 GENERAL TERMS, ABBREVIATIONS AND ACRONYMS

Table 11 : General Terms, Abbreviation and Acronyms

BCWS	Budgeted Cost of Work Scheduled
CBB	Contract Budget Baseline
CCB	Configuration Control Board
CDR	Critical Design Review
CENELEC	European Committee for Electrotechnical Standardization
CHARM	Change and Request Management
CR	Change Request
DDQS	Design, Develop and Qualify the Solution
DWF	Digital Web Frontend of ClearCase Document Management
EDM	ELEKTRA Delivery Model
EV	Earned Value
FEA-ID	Feature Identifier from Feature List
FQR	Final Qualification Review
GEN	Generic
HW	Hardware
IFS	Interfaces
IVV	Integration, Verification and Validation
KOM	Kick of Meeting
M	Mandatory
MPP	Manage Product Policy
N/A	Not applicable
O	Optional
PCM	Primavera Core Model
PCR	Problem Change Request
PDR	Preliminary Design Review
PEP	Personnel Efforts Planning
PiO	Put in Operation
PMP	Project Management Plan
QMP	Quality Management Plan
RAMS	Reliability, Availability, Maintainability and Safety
SE	Solution Element

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SEMP	System Engineering Management Plan
SFR	Solution Functional Review
SIL	Safety Integrity Level
SOR	Solution Orientation Review
SRR	Solution Requirement Review
SW	Software
TQR	Technical Qualification Review
TRR	Test Readiness Review
WBS	Work Breakdown Structure
WP	Work Package
WPD	Work Package Description

2.4.3 PROJECT SPECIFIC TERMS, ABBREVIATIONS AND ACRONYMS

Table 12 : Specific Terms, Abbreviation and Acronyms

ADM	Administration Diagnostic Monitoring
AzLM	Multi Section Axle Counter (one type of TDS)
BI	Basic Integrity
B-Vi	Basic Virtualization
CFI	Customer Furnished Item
DC	Data Centre
DCA	Distributed Controller Architecture
DDSP	Daten Diagnose Service Plattform (CFI ÖBB)
DFE	Digital Field Elements
DGP	Diagnosis Processor (ELEKTRA)
DIL	Digital Interlocking = DSTW
DSTW	Digitales Stellwerk = DIL
DSTW-ZE	Digitales Stellwerk Zentraleinheit = EIL
ECMUX	Element Controller MultipleXer
EIL	Electronic Interlocking System according to EULYNX – equals to DSTW-ZE
EKSA	Eisenbahnkreuzungssicherungsanlage = LCSI (CFI ÖBB)
EULYNX	European Initiative to Linking Interlocking Systems
FES	Field Element Station
GA DIL ÖBB	Generic Application Digital Interlocking ÖBB

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IEC	International Electrotechnical Commission
IPS	Integrated Platform Services (Integrierte Plattform Sicherungstechnik)
IPSS	IPS Solution Element
IXL	Interlocking
LCSI	Level Crossing Safety Installation = EKSA
LX	Level Crossing
MDM	EULYNX, Maintenance and Data Management
ÖBB	Österreichische Bundes Bahn (Austrian Railway Duty Holder, ÖBB)
OC	Object Controller
OMC	Operation Management Center = BFZ (Betriebsführungszentrale)
PZB	Punktförmige Zugbeeinflussung = Intermittent Train Control System
RaSTA	Rail Safe Transport Application (safety protocol for railway signalling systems)
RBC	Radio Block Center (CFI ÖBB)
RCU	Remote Controller Unit
SCI	Standard Communication Interface (EULYNX interface for process data information)
SCWS	Signal Controlled Warning System
SDI	Standard Diagnostic Interface (EULYNX interface for diagnostics)
SiDiNet	SicherungsDienstNetzwerk (CFI ÖBB)
SMI	Standard Maintenance Interface (EULYNX interface for maintenance)
Solution	A consistent set of systems, equipment and services provided to the Customer to meet a specific requirement
Solution Element	A system, a sub-system, a product / equipment, or a service, a software component, a hardware component
SSI	Standard Security Interface (EULYNX interface for security)
SSP	Security Services Platform (CFI ÖBB)
TDS	Train detection system
TMS	Traffic Management System
VIL	Virtual Interlocking
VSTW	Virtuelles Stellwerk = VIL

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3. SOLUTION ENGINEERING AND DEVELOPMENT PROCESS

The Solution will be conducted according to Management System 2025-08. As the Management System is continuously improving, it has to be mentioned, that in the evolution of the Solution, for newly created documents within this project, the respective valid templates have to be used according to the up-to-date Management System 2025-08 version. For maintenance and extension of existing documents the previously used template may remain.

3.1 ORGANIZATION OF SOLUTION ENGINEERING

The description of the roles mentioned below can be found on the Management System Portal.

3.1.1 SOLUTION CORE TEAM

Table 13: Solution Core team

Role	Resource	Deputy Resource
PM	H. Wolfram	
PDA	M. Eichberger	
PLM	R. Holzknecht	M. Dum
SEM	E. Stemmer	M. Gartner-Pohl
IVM	NN	
VER	see **) comment below	
SAM	C. Sünder	R. Büger
QAM	F. Rotter	A. Haslinger
VAL	P. Satlek	I. Grgic
RM	G. Eber	I. Miler
PSM	A. Willenberg	
WPM VIL	P. Bargiel	B. König
WPM IPSS	V. Vasiliev	
WPM DFE	B. Mandl	M. Goldenits

**) The verification tasks will be handled according to CENELEC [S2] EN 50716:2023 – Railway application – Requirements for software development. The review of all documents is performed through reviews by the roles defined in the Document List and Review Matrix for engineering projects [S23]. For details please refer to chapter 5.4.2 Role of Verifier (VER) according CENELEC EN 50716 and EN 50129

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3.1.2 SOLUTION TEAM

Table 14: Solution Team

Role	Resource	Deputy Resource
PLA	A. Paul	
ARC	Architectural Expert team from Solution Element coordinated by PDA E. Glaser M. Franzl M. Paslariu R. Holzmann E. Brenner K. Aichner	
CM	E. Rahmani	J. Cutig
SecAM	NN (successor of C. Lutze)	
DM	see *) comment below	n/a
RAM-AM	F. Moser	M. Scheuer
Integration	n/a	n/a

*) The role DM is not executed by a single person. For details please refer to chapter 5.4.3 Role of DM

3.1.3 EXTERNAL PROJECT ROLES

Table 15: External project roles

Role	Resource	Deputy Resource
Safety Assessor	Arsenal Race	n/a

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3.1.4 CONFIGURATION CONTROL BOARD (CCB)

Table 16: CCB

Role	Resource	Comment	M/O
PLM	see Table 13	Budget, schedule, scope	M
PM	see Table 13		M
SEM	see Table 13		M
PDA	see Table 13		M
SAM	see Table 13		M
IVM	see Table 13		O
VAL	see Table 13		M
QAM	see Table 13		M
Head of Engineering	M. Vana	for priority conflicts regarding resources in E	O

Other project team members are invited to the CCB as needed.

3.1.5 SUBCONTRACTORS

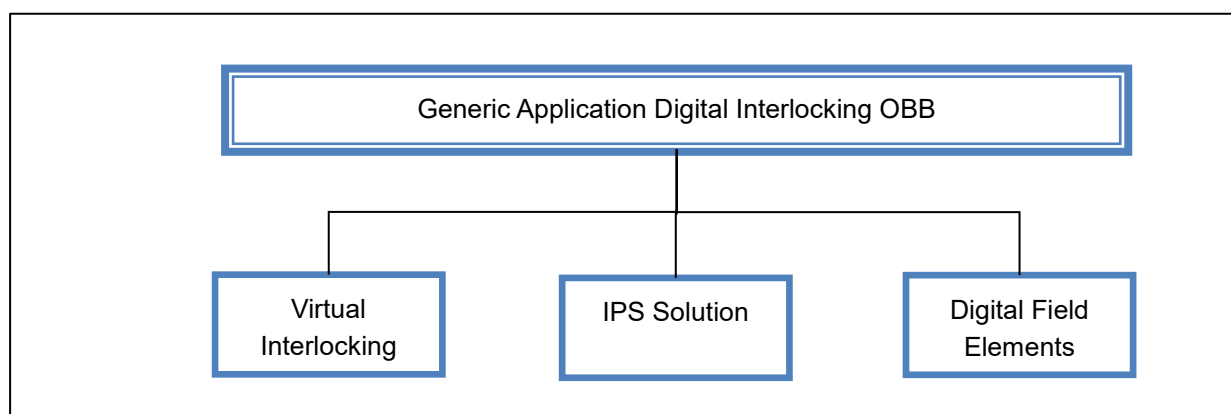
Not applicable in this solution.

3.2 SYSTEM ENGINEERING SUPPLIES

The Generic Application is according CENELEC 50129 responsible for the integration of several generic products and applications which are intended for use in digital interlocking installations on Austrian railway lines.

The Generic Application integrates a set of new solutions (including several products) as shown below. The Generic Application will integrate, test, verify, validate and assess the solution for further specific application in different Austrian operative projects.

Figure 3: Solution Overview Generic Application Digital Interlocking OBB (DSTW)



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3.3 INPUT DATA

The Generic Application is based in the input data of the Solution Elements. For Phase 1 following is defined as identified in Table 3: WBS managed as Solution

Table 17: Suppliers Input Data

Item	Version	Provider
Virtual Interlocking	VIL OBB 1.0.0	Hitachi Rail Austria
IPS Solution	IPSS OBB 1.0.0	Hitachi Rail Austria
Digital Field Elements	DFE 1.0.0	Hitachi Rail Austria
Interface Converter	IFC 1.0.0	Hitachi Rail Austria

3.4 PRECONDITIONS / RELATED SOLUTIONS

Table 18: Preconditions

Deliverable	Completion Date	Provider / comment
DDSP Itf-Specification ready	Phase 2 only	ÖBB
DDSP Platform ready for integration test ¹⁴	Phase 2 only	ÖBB
Refer to Solution Elements ¹⁵		

3.5 TECHNICAL OBJECTIVES

The requirements basis for the Generic Application is the „ÖBB_RA-DSTW-D1_Leistungsbeschreibung“.

¹⁴ This is at least related for the functionality to be provided in Phase 1

¹⁵ Preconditions are handled in the respective solution elements as defined in Table 3

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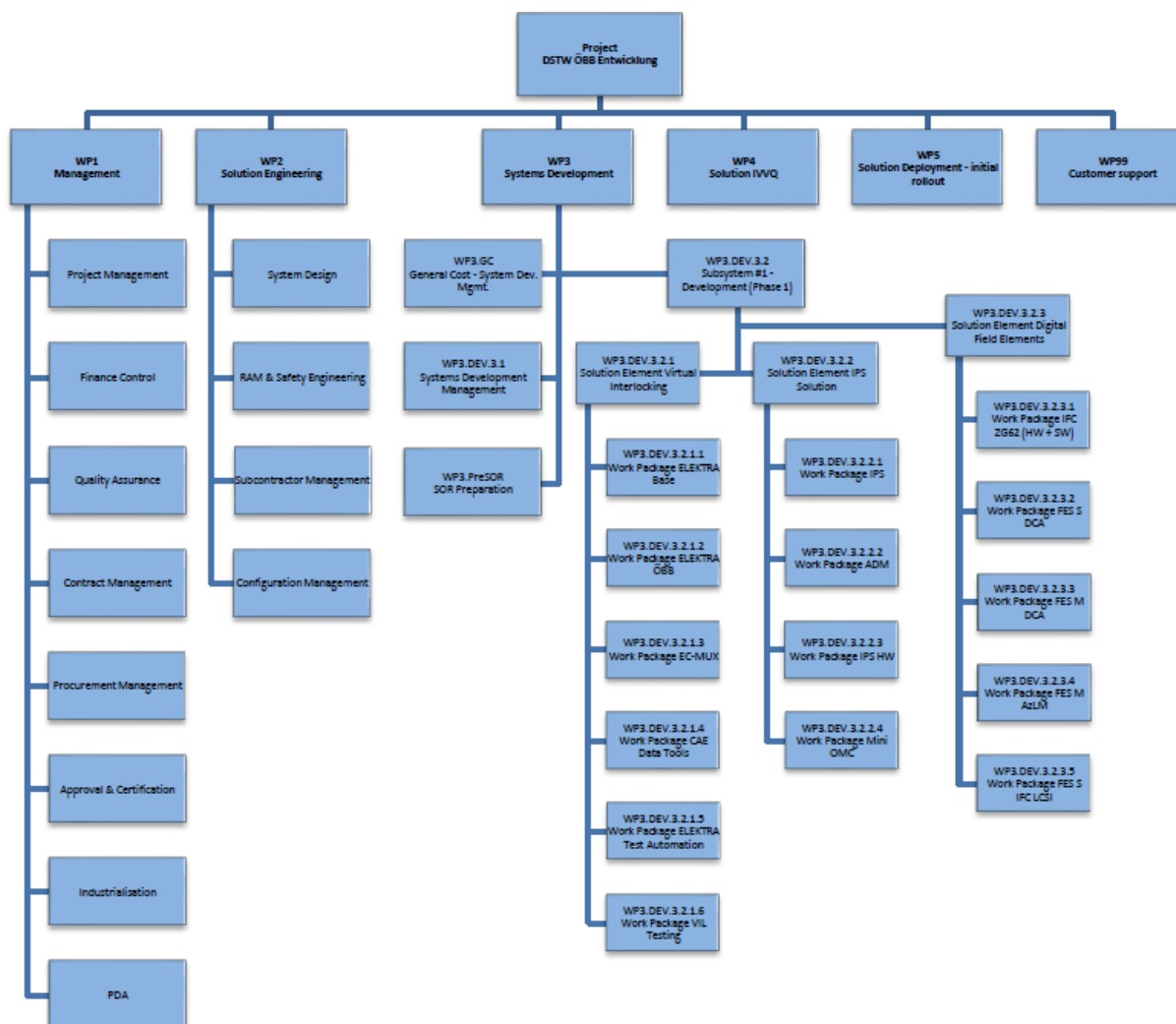
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3.6 DETAILED FIRST LEVEL WBS FOR WORK PACKAGES

Table 19: Detailed WBS



3.7 SPECIAL NEED FOR TRAINING OF SYSTEM ENGINEERING TEAM

A training plan will be set up:

- One focus will be on PM certification and also IVM certification
- introduction to new technologies (server) and new interfaces (EULYNX and System Pillar)
- safety principles of DSTW/DIL
- new concepts developed in the project, e.g. IPS background for all team members

3.8 CONSTRAINTS / RISKS

Constraints which are not already considered in the [RiskReport] are listed here.

a) None.

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3.9 WORK CONTAINED

Please refer also to general scope statement in section 1.1.1

- a) Engineering of the scope mentioned in section 1 according to the applying parts of the relevant standards listed in section 2.3.
- b) Delivery of the baseline documentation, including the Safety Cases.
- c) Delivery of the items listed in section 1.3.2. and table Table 6
- d) Coordination and execution of Factory Acceptance Tests (Customer labor-/integration- tests), and if a resolution of detected item during FAT is part of this project will be defined in the evaluation meeting. I.e. if not blocking then item can be moved to future baselines.
- e) Assessment Certification for this solution. Details have to be documented in Section 5.1.5 Validate the solution.

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3.9.1 WORK CONTAINED IN PHASE1 - BASIC VIRTUALIZATION (B-Vi)

Greenfield stations only

a) Digital Interlocking Central Unit / IPS-Solution:

- Hot-Hot Redundancy 2x2oo2 for ELEKTRA Core and 1oo2 for ECMUX based on TAS Platform 2.6.x and IPS-Solution
- Based on ÖBB BL 29.40 (extended with "Schutzweg Freiprüfung")
- "Static" Server Virtualization for the first stations based on OMC-3.0 solution. (Server Hardware provided by Hitachi - Network components and cabinets provided either by ÖBB or Hitachi) Relatively static Setup- und Administration of the system
- Adaption of ELEKTRA-Core and ECMUX for virtualization
- Integration of EULYNX BL4R4 compliant Object Controllers and required field elements.
- Basic Diagnosis and Administration via ADM (temporary SDI and remote download via ADM based on PULSE solution)
- Initial lab and test environment
- Support DDSP Design Phase or Initial DDSP Integration support if available in time
- Data Preparation tool (OREST, DMT, DCA Tooling)
- Test automatization
- Mini-OMC to replace EBO2 in the test environment
- time synchronization via NTP
- ECMUX - Point Movement Sequencer for limiting power supply (Weichenlaufkette)
- Day-Night switchover
- IFC-ZG62: Interface Converter ZG62 based on ÖBB-pilot with DCA 2.x technology
- EKSA implementation on ECMUX (for early start to allow solution in Phase 2)
- Cybersecurity implementation package 1 (Security Level 2 included in Security Case)
-> target to implement System Pillar (analysis ongoing on split Phase 1 and Phase 2)
ÖBB requests to be considered for early ability to test end-end security
- a. VIL feature set Phase 1
 - * SSI catalogue (telemetry, SCI, SDI) – please note that SCI is only implemented between the ECMUX and the OC
 - * certificate handling available - but no secured connection to OEC (only for testing purposes along implementation in DCA3)
 - * ECMUX (on secured handling limitation) - TLS switchable (on/off)
- b. IPS feature set Phase 1
 - * key generation (private keys)
 - * SSI catalogue (proxy, NAT, SCI, SDI for testing purposes)
- c. DFE feature set Phase 1 (along DCA features):
 - * test of DFE subsystem and components
 - * TAS PLF 2.6.0 cybersecurity upgrades
 - * TLS switchable (on/off) (OCS 3.5.2 has requirements to have configurable TLS cipher suites (as per the EULYNX technical specification. OCS shall support Ciphers to be adjustable. Key exchange methods shall also be configurable)
 - * Discussed priority with DCA team:
 - SCI R4R4 - SCI related functions focused (as precondition for ÖBB specific implementation)
 - ÖBB specific implementation
 - TAS-Platform Integration - for SDI+SMI+SSI+SCI-hashing (TLS) and Certification enrolment
 - Diagnose + Maintenance

b) DFE - FES:

- FES stations of the sizes S+M (FES-M AzLM, FES-S DCA and FES-M DCA)

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- Integration of DCA Release 1 (3.1.0)
Discussed priority with DCA team:
 - SCI R4R4 - SCI related functions focused (as precondition for ÖBB specific implementation)
 - ÖBB specific implementation
 - TAS-Platform Integration - for SDI+SSI+SCI-hashing (TLS) and Certification enrolment
 - Diagnose + Maintenance (remote update for DCA only)
- AzLM (TDS) v7.10 with EULYNX BL4R4 - target to be compliant with LFC10 / encryption (SDI to be limited)
- Subsystem Point
- Subsystem Light Signal (incl. Indusi Extender Box)
- Generic I/O
 - Dispatch Lamp,
 - Station and Line KeyLock,
 - UnivElement
- Lab and test environment setup
- Intermediate steps level crossing converter - part of SCI -> early start with availability in Phase 2

3.9.2 OUTLOOK PHASE 2 OF B-VI:

c) Digital Interlocking Central Unit / IPS Solution:

- Adaption of Recovery mechanisms and further improvements
- Additional CMOS data recovery in case of Safety Shutdown
- Flexible and scalable Server Virtualization
- Flexible Setup- and Administration via ADM and DDSP
- basic diagnosis and maintenance functions (depending on DDSP status)
- Full Cybersecurity measures based on System Pillar
- Integration of new DCA and AzLM Versions
 - New CPU LFC20 und CyberSecurity (LFC20) for DCA and corrections
 - Remaining AzLM features (SMI+SDI) and corrections
- Lab and test environment extensions
- DDSP (CFI) Diagnosis and full Security and encryption (depending on DDSP status)
- Maintenance mode for field elements (Betrieb mit tlw. außer Betrieb befindlicher Außenanlage)
- Axle counter reset for multiple sections
- EKSA implementation on ECMUX finalized
- Design need to support Geo-Redundancy – feature to be ordered by ÖBB

d) DFE - FES:

- FES of sizes L+XL (including 19" indoor racks)
- Integration of final DCA Release 3.1.2 (SMI)
- Integration of final AzLM Version v7.11 (SMI)
- Integration of LCSi Converter (SCI-LX) – full EKSA support
- IFC-SbL60A
- Lab and test environment setup
- Potential integration of updated & cost improved DCA devices (1-SEUC ICB) and 2-SEU ICB solutions. According to DCA team am availability by end 2027 is foreseen

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3.9.3 OUTLOOK PHASE 3 OF B-VI: REMAINING FUNCTIONAL FEATURES

e) Digital Interlocking Central Unit / IPS-Solution:

- Corrections if any
- SCI-ILS (neighbor VSTW, neighbor country)
- ECMUX logic for VLSA (Verkehrslichtzeichenanlage), if required
- OREST - Automatic ÖBB data import (rough estimation included; in offer but no detailed ÖBB inputs available yet)
- Local Point sector feature (Weichensektoralfreigabe) - ÖBB required solution via an App is not included in estimation and offer
- Full DDSP support (SMI included including remote-update)
- Nahbedienung / local control via OC/IO
- Self check feature for OCs and DIL
- Software activation via OMC

f) DFE - FES:

- FES XXL (tbc)
- New functionalities, corrections
- Selfcheck and System-Data-Activation for OCs
- FES for tunnel
- Generic I/O
 - Local Control
- Retrofit
 - Support of Siemens Outdoor facilities (*unclear answer in Bieteranfragen*)

3.10 WORK NOT CONTAINED

While the work contained is already stated in 3.9, the following topics are explicitly mentioned as not part of this solution in order to sharpen the responsibility – however tasks as integration and customer acceptance will be managed in the respective work-packages as described in section 1.1.3:

- a) Any HW, software, product or solution development not mentioned in chapter 3.9 (see solution elements. At this stage explicitly, the Development of a new iLED or iAnzeiger is excluded from this project) -> however the integration of new iLED/iAnzeiger is in scope of the project
- b) Industrialization of HW components (see solution elements) – *this need to be discussed in the project – added to open points*
- c) Execution of any field tests on site
- d) Execution of customer acceptance tests (details see [IVP] and WP description WP4)
- e) Coordination and execution of Factory Acceptance Tests (Customer lab-/integration-tests; (details see [IVP] and WP description WP4))
- f) Installation and put into operation at customer site
- g) Additional features as offered to ÖBB like Geo-redundancy, E-Shu, ...
-> however from actual tender Georedundancy is requested by the so-called step 3 from ÖBB – therefore preparation has to be done for Georedundancy in Phase 2 from Hitachi DIL to fulfil the timelines even it need to be ordered as additional feature.
- h) The integration of third-party OCs – however the integration activity is a clear request of the customer and can be requested by a separated purchase order

3.11 RESPONSIBILITY ASSIGNMENT MATRIX

The responsibilities of the roles at a Milestone Review can be seen in Milestone Template Solutions [S22].

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The milestone review TQR2 confirms the successful completion of product validation and the readiness of all needed elements for customer acceptance and that the Solution is ready for Assessment according to CENELEC. The PDA confirms with the approval to milestone TQR2 that the delivered solution meets his/her expectations.

The milestone review FQR confirms that the Design, Development and Qualification activities including Assessment according to CENELEC are complete and that the Solution is ready for production, deployment and transition to support.

3.12 TOOLS AND FACILITIES

The following subsections provide a rough overview of the project tool setup. For version information and tool classification see document [ToolClassSW].

3.12.1 MANAGEMENT AND DOCUMENTATION TOOLS

Table 20: Management and Documentation Tools

Tool Category	Tool	Note
Project Management	Primavera PCM	
Task Management	Jira itask	
Requirements Management	DOORS/RMF, Polarion	DOORS/RMF as customer defined tool
Management Traceability	DOORS/RMF, Polarion	DOORS/RMF as customer defined tool
Configuration Management	ClearCase Git	
Change Management	CHARM	
Documentation Management	ClearCase DWF SharePoint	
Documentation Generation	DOORS/RMF Polarion MS Office	
IVV Management	Polarion MS Excel	

3.12.2 CODE BUILDING TOOLS

NA ¹⁶

¹⁶ Code building tools are handled in the respective solution elements as defined in Table 3

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4. PROCESS CONTROL

4.1 CONTROL OF SOLUTION DEVELOPMENT

The solution elements and assigned responsibilities are listed in Table 3: WBS managed as Solution.

4.1.1 DEVIATION

The development of a Solution has to be planned and executed according to the DDQS Processes of Management System. Deviation to the description in Project Life Cycle [S20] is described in section 5.

4.1.2 METRICS

The used metrics for control of the engineering process are identified in the Engineering Dashboard [EngDB].

The tailoring for this dashboard is listed below:

Table 21: Metrics

Sheet	Tailoring
☒ Actions & Decisions	Used - no deviation.
☐ Stability of Team Velocity	Not used. See Note 5 below.
☐ Visibility Point Burnup	Not used. See Note 5 below.
☐ Code Quality	Not used.
☐ Code Reuse	Not used.
☐ Defect management	Not used.
☒ Documentation	Used – only System Level documents reported
☒ Finance	Used. See Note 1 below.
☐ Interface testing progress	Not used.
☐ IVV Test Writing	Not used.
☒ Master Plan	Used - no deviation.
☒ Milestones	Used - no deviation.
☒ OBS	Used - no deviation.
☒ PCR (Problem Change Request)	Used. See Note 2 below.
☒ Quality	Used. Data provided by QAM.
☐ RAMS	Optional. Data provided by RAM-AM and SAM.
☐ Release content progress	Optional. Not only features are tracked, but also legacy PCRs which are part of the scope as defined in the Fault List
☐ Remaining work burndown	Not used.
☒ Requirement Analysis	Used. See Note 3 below.
☐ Requirement Flowdown	Not used.
☐ Sizing & critical performance	Not used.
☒ Technical Risks	Used - no deviation.
☐ Validate & qualify test progress	Not used.
☒ Verification testing progress	Used. See Note 4 below.
☒ WBS	Used - no deviation.
☒ Workload	Used - no deviation.

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Note 1: ⓘ Guideline:

- a) "BCWS Planned Spend" is the monthly planned costs of work. This value can be determined from PCM. It is the defined/updated at the time the CBB is defined or updated.
- b) "Forecast" is the spending forecast from the last month with actuals to the end of the project (before this period it has to be empty).
- c) "EV" is not used.

Note 2: ⓘ Guideline:

This metric deals with PCRs found in the phase "Integrate and Verify the Solution". Hence no baseline can be given ahead. The "Raised Forecast" may be predicted after a few months of this phase from past data.

Note 3: ⓘ Guideline:

Only the table "requirements chart data" is filled out, were the

- d) "Total number of Req.": is the sum of all approved requirements at the time of the reporting. The reporting is done every month
- e) "Baseline": a new baseline is created, when a new release of the SSS is done and a new release label is generated in DOORS
- f) "Analysis forecast": specifies the sum of current and new requirements.
- g) Either a value in "Baseline" or in the "Analyses forecast" is filled out except in the current month, where both fields are filled out.

Additionally, to be filled out are:

- "Analyzed Req.": sum of all requirements, which are analyzed and not changed
- "Req. to be clarified": sum of requirements, which are currently not approved, new or changed, may not be the sum of the below attributes
- "Obsolete Req.": sum of requirements, which will be set to descoped
- "Modified Req.": sum of changed requirements
- "New Req.": sum of new requirements

Requirements characterization can be generated via DOORS, via the menu "TRSS-AT" -> "Simple Functions" -> "KPI" -> "Input for Req. analysis tab".

Note 4: ⓘ Guideline:

Field "Req. conformity (%)" may be used as a percentage of tested and verified items of the scope defined in [SoC].

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Note 5: ⓘ Guideline:

The monitoring of project progress within an agile development environment requires a set of metrics defined in the Agile Definition SMP [S25] document sections 9.2.1 'Stability of Team Velocity' and 9.2.2 'Visibility Point Burnup' and is expected by senior management to be displayed during the PPPRs.

Furthermore, the tailoring of the monitoring of project progress within an agile development environment in chapter 5.5 must be taken into consideration to get an adequate information basis for the KPIs.

4.2 MONITORING AND CONTROL OF SOLUTION ENGINEERING ACTIVITIES

4.2.1 PERIODIC INTERNAL REVIEWS AND MEETINGS

Table 22: Reviews and Meetings

Meeting	Responsible	Attendees	Frequency	Objectives
periodically Solution Core Team Meeting	SEM	"Solution core team", ARC, WPM(s) Only on demand: CM	weekly	Project status and action points.
SEM alignment	SEM	SEM, IVM and SEMs of Solution Elements	weekly	Alignment on solution level on timeline, deliverables, dependencies
Risk and opportunity review	SEM	PM, SEM, PLM, PDA, IVM, SAM, ARC, QAM	Quarterly or on demand	
Resource review	SEM	PM, SEM, HoD	monthly	Check utilisation and additional needs of solution team.
Cost tracking	Project controller	PM, SEM, HoD	monthly	Report of costs and progress to Financial Control Director.
Periodically Performance Project Review (PPPR)	SEM	PM, SEM, QAM	monthly	Report of costs and progress, KPI dashboard, Progress Matrix, required decisions from SM to HoD Project Engineering, HoD Engineering

4.2.2 MILESTONE REVIEWS AND AUDITS

Milestone reviews are done according to Milestone Template Solutions [S22]. Quality Audits will be conducted through the QAM – see [QMP]. Safety Audits will be conducted through the SAM – see [PSP].

4.2.3 REPORTING

Reporting is done via the Engineering Dashboard [EngDB].

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4.3 CRITICAL TECHNOLOGIES

Not applicable in this solution.

4.4 RISK AND OPPORTUNITIES CONTROL

The Risk and Opportunities Control in this Solution will follow the Management System Process Manage Risks and Opportunities [S7] and is monitored in the [RiskReport].

4.5 CORRECTIVE ACTION PROCESS

Information about the Corrective Action Process can be found in the [QMP].

4.6 INDEPENDENT EVALUATION AND VERIFICATION

Independent evaluation and verification will be done by

1. the verifier and domain experts upon instruction of the verifier (VER);
2. the validator (VAL);
3. the safety assurance manager (SAM).

4.7 ESTIMATE/RE-ESTIMATE

The initial estimation is either an input for the phase “Orientation the Solution” or is done or verified during this phase. Re-estimation of remaining work is done during the progress of working packages typically on a monthly period. The method used for estimation is mainly “Expert estimations” based on past experience, educated assumptions and risk considerations.

4.8 PLANNING

Planning is done in the tool PCM and can be found there with the identification listed in Table 1.

4.9 VALUE ANALYSIS AND DESIGN TO COST

Value Analysis and Design to Cost is done by the Engineering Dashboard [EngDB].

4.10 PEER REVIEWS

The Peer Reviews in this Solution will follow the Minimum Requirements for the configuration management implementation [S9]. An overview of mandatory Documents and Reviewer has to be done according to Document List and Review Matrix for engineering projects [S23].

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5. DEVELOPMENT SCENARIO AND TAILORING OF THE SYSTEM ENGINEERING PROCESS

This development process follows the initialisation by Management System process Manage Product Policy (MPP) as described in Project Life Cycle [S20].

Figure 4: Develop and Qualify the Solution (DDQS)

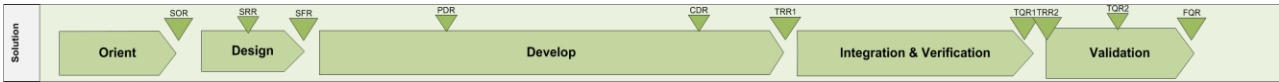
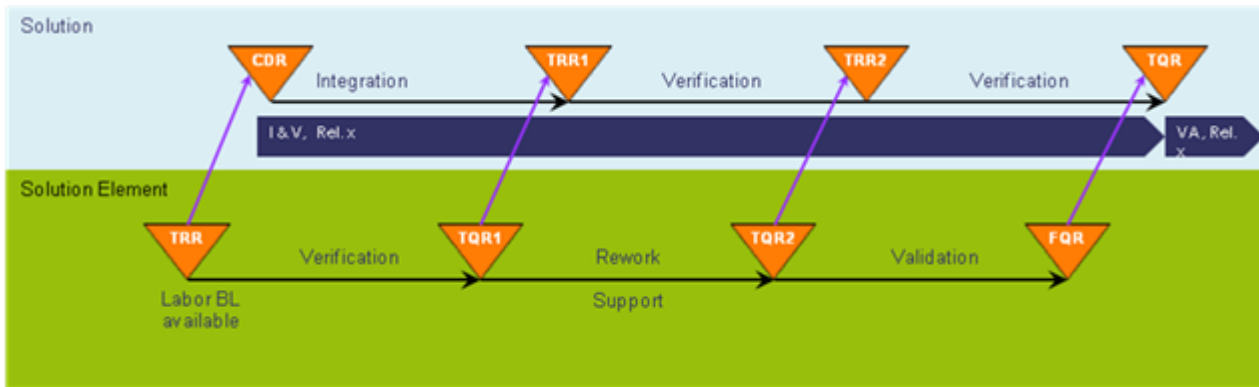


Figure 5: Interaction of solution and solution elements



5.1 TAILORING FOR THIS SOLUTION (DDQS PHASES)

5.1.1 ORIENT THE SOLUTION

No other deviation to the description in Project Life Cycle [S20].

5.1.2 DESIGN THE SOLUTION

No deviation to the description in Project Life Cycle [S20].

5.1.3 DEVELOP THE SOLUTION

No deviation to the description in Project Life Cycle [S20].

5.1.4 INTEGRATE AND VERIFY THE SOLUTION

No deviation to the description in Project Life Cycle [S20].

5.1.5 VALIDATE THE SOLUTION

The Assessment Certification according to CENELEC is needed for this solution. No other deviation to the description in Project Life Cycle [S20].

5.2 TAILORING OF DOCUMENTATION MANAGEMENT

The tailoring of the Product Line Engineering (PLE) is used as described in Project Life Cycle [S20].

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The naming convention for the name “Firstname Lastname” of the author, verified by, approver of a document can also be “F. Lastname”.

No other deviation to the description in Document Management Rules [S21].

5.3 TAILORING OF MILESTONES

No deviation to the description in Project Life Cycle [S20] and Agile Definition SMP [S25].

5.4 TAILORING OF ROLE ASSIGNMENTS

5.4.1 ROLE OF IVVQM

The role IVVQM is not executed by a single person. The major IVVQM activities are distributed to other roles:

- Integration: is made by IVM
- Verification: is made by IVM
- Validation: is made by VAL
- Qualification: is made by VAL (in case of HW development)

5.4.2 ROLE OF VERIFIER (VER) ACCORDING CENELEC EN 50716 AND EN 50129

The verification tasks will be handled according to CENELEC [S2] EN 50716:2023 – Railway application – Requirements for software development. The review of all documents is performed through reviews by the roles defined in the Document List and Review Matrix for engineering projects [S23]. The role VER is not executed by a single person. The responsibilities are defined in Table B.6 in [S2] and Table G.3 in [S6] (EN 50126-2:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 2: Systems Approach to Safety) :

- Process Verification: is made by QAM
- Final document Verification: is made by SEM or a person assigned by the SEM within the [DoP]
- Software/Code Verification: NA
- Deliverables Verification: is made by IVM
- Enabling systems Verification: is made by CM
- Lifecycle phases Verification: is made by QAM
- Collection of Verification results in Verification-report: is made by IVM or a person assigned by the SEM within the [DoP]

5.4.3 ROLE OF DM

The role DM is not executed by a single person. The major DM activities are distributed to other roles:

- new ANV: can be obtained by SEM
- Document Index: is made by CM or SEM
- Formal check at release of a document: is made by QAM
- Other DM activities: are made by author of the document

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5.5 TAILORING OF THE MONITORING OF PROJECT PROGRESS WITHIN AN AGILE DEVELOPMENT ENVIRONMENT

Remark: No development is planned by this Generic Application, therefor this tailoring is not used.

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6. SYSTEM ENGINEERING SUPPORT

6.1 DATA AND CONFIGURATION MANAGEMENT

Data and Configuration Management is carried out according to Management System Process 9.7 CFG - Manage Configuration [S15], Document Management Rules [S21] and the specific Configuration Management Plan [CMP].

6.2 CONFIDENTIALITY

The rules as listed in Protection of Group Information [S18] apply.

6.3 EXPORT CONTROL

Export Control is carried out according to Management System Process 9.3 Control Export [S16].

6.4 CAPITALIZATION

Capitalization is handled in the context of the related Manage Product Policy process.

6.5 MEASUREMENTS AND ANALYSIS

See section 4.

6.6 TECHNICAL AND NON-TECHNICAL REQUIREMENTS MANAGEMENT AND CONTROL

For Requirement Management the tools DOORS and Polarion are used. The Requirements Manager ensures that requirements are

- uniquely identified
- unambiguous, testable and consistent
- traced from origin through all systems
- have all changes managed

The Requirements Manager manages changes throughout the life of the project. If requirements change, the impact will be reflected in the project plans, work products, and activities. Any inconsistencies between these customer- and/or system requirements and project plans, work products, and activities will be identified and resolved.

6.7 VALIDATION OF REQUIREMENTS

The validation is done in accordance with the Validation Qualification plans as listed in Documentation Plan [DoP].

6.8 DECISION ANALYSIS AND RESOLUTION

Decision analysis and resolutions are part of design and review processes. The results are documented in design documents, review reports and in this document.

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6.9 SUBORDINATE PLANS

Subordinate plans according to Table 3: WBS managed as Solution are listed in Documentation Plan [DoP].

6.10 QUALITY ASSURANCE

The Quality Assurance in this Solution will follow Management System Guarantee quality and manage continuous improvement [S17]. Additional information can be found in the Quality Management plans as listed in Documentation Plan [DoP].

6.11 CERTIFICATION SUPPORT ACTIVITY

See Safety Management plans as listed in Documentation Plan [DoP].

6.12 HSE PLAN

The local requirements of the HSE Management and the measures resulting from it for the environment and protection of the employee’s health are described in the HSE Instruction [S24]. HSE Plan is listed in Documentation Plan [DoP].

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APPENDIX A CRITERIA FOR THE SELECTION OF TECHNIQUES
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Solution designed according to EN50716 [S2]. Will be updated during second SOR (see Table 5: Milestones Generic Application Digital Interlocking OBB 1.0.0).

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APPENDIX B REQUIREMENTS OVERVIEW DIGITAL
INTERLOCKING ÖBB

In this appendix an overview on all requirements and specification documents including their dependencies is listed. This includes documents which are created in the context of the digital interlocking project on Generic Application Level and all three solution elements.

The list reflects the status during preparation of the SRR-1 milestone, means prior to contract signature.

Therefore, the status is work in progress and the document is available on SharePoint:

[DIL_requirements_overview.xlsx](#)

As soon maturity is reached and latest with SOR-2 the list will be included to ClearCase under version control.

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DOORS	LH / CRS	System	Subsystem	Component	DocID	Title
						2c_ÖBB_Ausschreibungsunterlagen_4.Runde
					3BU_17300_0001_DSAPC.04	Customer_Requirements_Specification DSTW Streckenausrüstung OBB - Teil A Verfahren
EL4 Rahmenvereinbarung					3BU_17300_0002_DSAPC.04	Customer_Requirements_Specification DSTW Streckenausrüstung OBB - Teil B Rahmenvereinbarung
					3BU_17300_0003_DSAPC.04	Customer_Requirements_Specification DSTW Streckenausrüstung OBB - Teil C Vertrag
					3BU_17300_0004_DSAPC.04	Customer_Requirements_Specification DSTW Streckenausrüstung OBB - Teil D Leistungsbeschreibung und Lastenhefte
						D1 Leistungsbeschreibung
EL4 LH D2						D2 Lastenhefte DSTW
						2025-05-23_OBB_RA-DSTW-D2_Lastenheft_DSTW_V1.3(ÄND).pdf
						2025-05-23_OBB_RA-DSTW-D2_Lastenheft_DSTW_V1.3.pdf
						D2.0_Glossar.pdf
						D2.1_Referenzliste_V1.3.pdf
--->						D2.2_Teillastenheft_RAMS_V1.3.pdf
--->						D2.3_Teillastenheft_Allgemeine_Anforderungen_V1.3.pdf
--->						D2.4_Teillastenheft_Security_V1.3.pdf
--->						D2.5_Teillastenheft_Teilsystem_DSTW-ZE_V1.3.pdf
--->						D2.6_Teillastenheft_Teilsystem_Weiche_V1.3.pdf
--->						D2.7_Teillastenheft_Teilsystem_Lichtsignal_V1.3.pdf
--->						D2.8_Teillastenheft_Externe_EKSA_V1.1.pdf

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--->						D2.9_Teillastenheft_Teilsystem_Generische_Ein_und_Ausgabe_V1.2.pdf
--->						D2.10_Teillastenheft_Teilsystem_Gleisfreimeldung_V1.3.pdf
--->						D2.11_Teillastenheft_Feld_Element_Station_V1.1.pdf
						D3 Leistungsverzeichnis
EL4 LH D4						D4 LH Rechenzentrum
EL4 LH D5						D5 LH EKSA-DSTW
EL4 LH D6						D6 LH Netze
EL4 LH D7						D7 LH ETCS L2plus
						D8 Security Konzept
						D9 IT-Inbetriebnahme Konzept
						D10 DSTW-SIL-Allokation
						D11 Konzept DDSP
						D12 Leistungsbeschreibung Service
						D13 Leistungsverzeichnis Service
					3BU_17300_0005_DSAPC.04	Customer_Requirements_Specification DSTW Streckenausrüstung OBB - Teil E Mitgeltende Dokumente
					3BU_17300_0006_DSAPC.04	Customer_Requirements_Specification DSTW Streckenausrüstung OBB - Teil F Beizubringende Unterlagen
						FL
EL4 FL					3BU_17300_0000_FLAPC	Feature List DSTW
EL-OBB - FL					3BU_15000_4301_FLAPC	Feature List Elektra ÖBB

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DOORS						SSS
					DocID	Title
EL4 SSS, EL4 IDJ-RA	D2.3 <---	DIL			3BU_17300_1251_DTAPA	System Requirements Specification - Digital Interlocking OBB (DIL) - General Requirements
EL4 SSS	D2.4 <---	DIL			3BU_17300_1252_DTAPA	System Segment Specification (SSS) - Cybersecurity DSTW ÖBB
EL4 SSS, EL4 IDJ-RA	D2.5 <---	DIL			3BU_17300_1255_DTAPA	System Requirements Specification - Digital Interlocking OBB (DIL) - Central Unit Part 1
EL4 SSS, EL4 IDJ-RA	D2.6/7/9/11 <---	DIL			3BU_17300_1256_DTAPA	System Requirements Specification - Digital Interlocking OBB (DIL) - Field Element Controller and Stations
EL4 SSS, EL4 IDJ-RA	D2.2 <---	DIL			3BU_17300_1257_DTAPA	System Requirements Specification - Digital Interlocking OBB (DIL) - RAMS
EL4 SSS, EL4 IDJ-RA					3BU_17300_1258_DTAPA	?
EL4 SSS, EL4 IDJ-RA					3BU_17300_1259_DTAPA	?
EL4 SSS					3BU_17300_125x_DTAPA	SUBSYSTEM REQUIREMENTS SPECIFICATION Digital Interlocking OBB SYSTEM Requirements Specification DIL INTERLOCKING (DIL)
EL4 SSS, EL4 IDJ-RA		DIL ???? <---	VIL		3BU_15000_6955_DTAPA	SUBSYSTEM REQUIREMENTS SPECIFICATION Digital Interlocking OBB Digital Interlocking OBB
?		DIL 1252 <---	VIL		3BU_15000_6980_DTAPA	SSS Security Subsystem VIL
?		DIL ???? <---	VIL 6955 <---	GenSys	3BU_15000_6801_DTAPA	Segment Requirements Specification Car-red ELEKTRA 4.0 for OBB
?		DIL ???? <---	VIL 6955 <---	GenSys	3BU_15000_6850_DTAPA	SSS Component Generic System
?		DIL ???? <---	VIL 6955 <---	GenSys	3BU_15000_0300_DTAPA	SRS Generic Elektra
?		DIL ???? <---	VIL 6955 <---	GenSys	3BU_15000_0315_DTAPA	SRS Generic Elektra ÖBB
?		DIL ???? <---	VIL 6955 <---	Appl	3BU_15000_4315_DTATA	SRS Betriebliche Funktionen

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?		DIL ????? <---	VIL 6955 <---	Appl	3BU_15000_4335_DTATA	SRS Betriebliche Funktionen
?		DIL ????? <---	VIL 6955 <---	Appl	3BU_15000_4377_DTAPA	System Requirement Specification - Test Automation For ELEKTRA
?		DIL ????? <---	VIL 6955 <---	Appl	3BU_15000_2116_DTATA	SRS Betriebliche Funktionen
?		DIL ????? <---	VIL 6955 <---	Appl	3BU_15000_4112_DTATA	SRS Betriebliche Funktionen
?		DIL ????? <---	VIL 6955 <---	ECMUX	3BU_61400_0200_DTAPA	System Specification SSS, ELEKTRA 4 - VIL-ECMUX
?		DIL ????? <---	VIL 6955 <---	ECMUX	3BU_61400_0136_DTAPA	SSS Point 3BU 61400 0136 DTAPA
?		DIL ????? <---	VIL 6955 <---	ECMUX	3BU_61400_0170_DTAPA	System Segment Specification SEA SW OBB - Dispatch Lamp/Eu1ynx
?		DIL ????? <---	VIL 6955 <---	ECMUX	3BU_61400_0174_DTAPA	System Segment Specification SEA SW GEN - Universal Element / Eulynx
?		DIL ????? <---	VIL 6955 <---	ECMUX	3BU_61400_0182_DTAPA	System Segment Specification - SEA SW OBB - Light Signals / Eulynx
?		DIL ????? <---	VIL 6955 <---	ECMUX	3BU_61400_0186_DTAPA	SSS Keylocks
?		DIL ????? <---	VIL 6955 <---	ECMUX	3BU_61400_0139_DTAPA	SSS Key Lock
?		DIL ????? <---	VIL 6955 <---	ECMUX	3BU_61400_0173_DTAPA	System Specification SSS, SEA SW - Axle Counter / Eulynx
?		DIL ????? <---	VIL 6955 <---	ECMUX	3BU_61400_0180_DTAPA	System Specification SSS, SEA SW- Level Crossing External/ Eulynx
?		DIL ????? <---	VIL 6955 <---	ECMUX	3BU_61400_0188_DTAPA	System Specification SSS, SEA SW - Local Control / Eulynx
?		DIL ????? <---	VIL 6955 <---	ECMUX	3BU_61400_0190_DTAPA	System Specification SSS, SEA SW - Power Supply
?		DIL ????? <---	VIL 6955 <---	ECMUX	3BU_61400_0200_DTAPA	System Specification SSS, ELEKTRA 4 - VIL-ECMUX
?		DIL ????? <---	VIL 6955 <---	CAE	3BU_15000_4603_DTAPC	SRS OREST - OREST Allgemeines
?		DIL ????? <---	VIL 6955 <---	CAE	3BU_15000_6488_DTAPQ	SRS OREST - OREST Allgemeines- Baulisten
?		DIL ????? <---	VIL 6955 <---	CAE	3BU_15000_4634_DTAPC	SRS OREST - OREST Funktionen
?		DIL ????? <---	VIL 6955 <---	CAE	3BU_15000_4633_DTAPC	SRS OREST- Arbeitsschritte Teil 2
?		DIL ????? <---	VIL 6955 <---	CAE	3BU_15000_4631_DTAPC	SRS OREST- Arbeitsschritte Teil 3
?		DIL ????? <---	VIL 6955 <---	CAE	3BU_15000_6432_DTAPC	SRS OREST- Arbeitsschritte Teil 4

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?		DIL ???? <---	VIL 6955 <---	CAE	3BU_15000_6433_DTAPC	SRS OREST- Arbeitsschritte Teil 5
EL4 SSS, EL4 DFE SSS, EL4 IDJ-RA		DIL 1256 <---	DFE	FES	3BU_62800_0001_DTAPA	SUBSYSTEM REQUIREMENTS SPECIFICATION Digital Interlocking OBB Field Element Stations (FES-S and FES-M)
EL4 SSS, EL4 DFE SSS, EL4 IDJ-RA		DIL 1256 <---	DFE 0001 <---	OC-x	3BU_62800_0010_DTAPA	SUBSYSTEM REQUIREMENTS SPECIFICATION Digital Interlocking OBB DCA – General Requirements
EL4 SSS, EL4 DFE SSS, EL4 IDJ-RA		DIL 1256 <---	DFE 0001 <---	OC-P	3BU_62800_0011_DTAPA	SUBSYSTEM REQUIREMENTS SPECIFICATION Digital Interlocking OBB DCA - Point
EL4 SSS, EL4 DFE SSS, EL4 IDJ-RA		DIL 1256 <---	DFE 0001 <---	OC-LS	3BU_62800_0012_DTAPA	SUBSYSTEM REQUIREMENTS SPECIFICATION Digital Interlocking OBB DCA - LS
EL4 SSS, EL4 DFE SSS, EL4 IDJ-RA		DIL 1256 <---	DFE 0001 <---	OC-IO	3BU_62800_0013_DTAPA	SUBSYSTEM REQUIREMENTS SPECIFICATION Digital Interlocking OBB DCA – I/O
EL4 SSS, EL4 DFE SSS, EL4 IDJ-RA		DIL 1256 <---	DFE 0001 <---	OC-TDS	3BU_62800_0020_DTAPA	SUBSYSTEM REQUIREMENTS SPECIFICATION Digital Interlocking OBB Train Detection System
EL4 SSS, EL4 DFE SSS, EL4 IDJ-RA		DIL 1256 <---	DFE 0001 <---	IFC-Block	3BU_62800_0030_DTAPA	SUBSYSTEM REQUIREMENTS SPECIFICATION Digital Interlocking OBB IFC - Block
EL4 SSS, EL4 DFE SSS, EL4 IDJ-RA		DIL 1256 <---	DFE 0001 <---	IFC-Block	3BU_62800_0031_DTAPA	Subsystem Requirements Specification - DFE - IFC-Block SW
EL4 SSS, EL4 DFE SSS, EL4 IDJ-RA		DIL 1256 <---	DFE 0001 <---	IFC-LCSI	3BU_62800_0050_DTAPA	SUBSYSTEM REQUIREMENTS SPECIFICATION Digital Interlocking OBB IFC - LCSI
EL4 SSS, EL4 DFE SSS, EL4 IDJ-RA		DIL 1256 <---	DFE 0001 <---	FES	3BU_62800_0070_DTAPA	Subsystem Requirements Specification - DFE - Security FES HW
EL4 SSS, EL4 DFE SSS, EL4 IDJ-RA		DIL 1256 <---	DFE 0001 <---	IFC-Block	3BU_62800_0071_DTAPA	Subsystem Requirements Specification - DFE - Security HW IFC Block HW
EL4 SSS, EL4 DFE SSS, EL4 IDJ-RA		DIL 1256 <---	DFE 0001 <---	IFC-Block	3BU_62800_0072_DTAPA	Subsystem Requirements Specification - DFE - Security IFC Block SW

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EL4 ADM SSS		DIL ???? <---	ADM		3BU_17300_2101_DTAPA	System Subsystem Specification ADM-Subsystem, Digital Interlocking OBB
EL4 IPS SSS		DIL ???? <---	IPS		3BU_17300_2051_DTAPA	System Subsystem Specification IPS-Subsystem, Digital Interlocking OBB
					?	IPS HW
					?	IPS Workflows
					?	Security IPS

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